EVALUATING LIMITS

Expanding to find the hidden factor

The initial form obscured the factor of $(x - 1)$; expansion made it visible.

01

FACTOR HIDDEN — BEFORE

$$\lim_{x \to 1} \frac{(x+1)^2 - 4}{x - 1}$$

↓ Expand the numerator — $(x + 1)^2 - 4 = x^2 + 2x + 1 - 4 = x^2 + 2x - 3$

02

EXPANDED

$$\lim_{x \to 1} \frac{x^2 + 2x - 3}{x - 1}$$

↓ Factor — $x^2 + 2x - 3 = (x - 1)(x + 3)$

03

SHARED FACTOR VISIBLE

$$\lim_{x \to 1} \frac{(x - 1)(x + 3)}{x - 1}$$

↓ Cancel (x - 1) — then substitute x = 1

04

SUBSTITUTE — AFTER

$$\lim_{x \to 1} (x + 3) = 4$$